JAKOB SPURGAT

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PROFESSIONAL SUMMARY

Mechanical and electrical engineer with eight years of automotive experience spanning product development, design validation, and test automation. Experienced in developing mechanical products from concept through production launch and integrating hardware and software into custom engineering systems. Able to combine hands-on design, experimental testing, and root cause analysis to improve product performance and engineering workflows.

TECHNICAL SKILLS

- **Product Development**: mechanism and electromechanical design, DFM, DFSS, FMEA, Agile PLM, root cause analysis, rapid prototyping
- **Engineering Analysis and Testing:** experimental design, test method development, design validation, simulation and modeling, data analysis
- **Systems and Automation:** Systems integration, embedded systems, control systems, robotics, instrumentation and data acquisition
- **Programming:** Python, VBA (Excel), C++, Java
- **CAD and Simulation:** NX, CATIA, Onshape, Inventor, Fusion 360, MSC Adams

EXPERIENCE

Chip Ganassi Racing | Marysville, OH **R&D Engineer** **July 2022-Present**

Provide on-site engineering support for a major automotive manufacturer, developing new tools and processes to improve R&D and product development workflows

- Automated data validation, saving hundreds of hours per year compared to manual checks, which allowed issues to be caught live during testing
- Developed a robot to automate legacy test equipment, removing the need for human involvement in a previously error-prone test process
- Built a custom OBD-II device to automatically restore prototype vehicles from limp mode, reducing downtime during testing
- Created a remote weather monitoring station and integrated it into existing data acquisition so weather corrections are automatically applied during post-processing
- Reverse engineer CAN UDS messages, including security handshakes, so data logging can be performed on competitor vehicles
- Reverse engineer LIN messages so detailed tests can be performed on individual components
- Developed a searchable database so legacy data can quickly be accessed
- Mentor junior engineers and technicians to expand their technical capabilities

Magna Seating | Novi, MI **Product Engineer** **July 2018-July 2022**

Was responsible for automotive seat frame and mechanism design, from initial design through launch

- Designed products to meet safety, cost, manufacturability, and customer requirements
- Supported product launches by investigating manufacturing issues, implementing corrective actions and process changes, and recommending design improvements for manufacturability
- Managed bills of materials (BOMs) and parts revisions in Agile PLM and submitted engineering changes for approval
- Developed simulations to estimate operating effort of proposed designs
- Functioned as company representative in meetings with automakers to identify root causes of warranty claims
- Coordinated design validation, tracked test schedules, investigated and documented issues, and communicated results to engineering and management
- Handled algorithm optimization during development of in-house DFSS software

Magna Seating | Novi, MI **R&D Engineering Co-op** **June 2013-July 2018**

- Designed and conducted experiments to quantify the effects of design parameters on product performance

EDUCATION

Kettering University | Flint, MI **Magna Cum Laude** **Graduated June 2018**

- Bachelor of Science, Mechanical Engineering
- Bachelor of Science, Electrical Engineering

LEADERSHIP/VOLUNTEER

FIRST Robotics **Lead Programming Mentor** **2017-Present**

- Mentor high school students in software development, control systems, and robotics
- Collaborate with fellow mentors to develop ways to accelerate robot development, expand testing time, and improve software and mechanism reliability under competition conditions
- Contribute to sustained improvements in team competitiveness, resulting in a World Championship finals appearance